

# Muhammad Arslan Tabish

**Full Stack AI Engineer** · Claude API · Agentic Pipelines · Python & FastAPI · React & TypeScript  
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**Nationality** Pakistani · **Location** Lahore, Pakistan · **Relocation** Open to relocation and remote · **Notice** 30 days

## Profile

Full Stack AI Engineer with 3 years of professional experience building LLM-powered products end to end, from Python and FastAPI services and data models through to the React 19 and TypeScript interfaces I design and ship myself. Focused on **AI systems, not AI demos**: resumable multi-stage agentic pipelines, schema-constrained output with validation, retry and repair, grounded generation and RAG, guardrails and automated QA gates, model routing and per-call cost accounting on the **Anthropic Claude API**. Owns features from PostgreSQL schema to browser and into production on Docker, Vercel and Railway.

## Technical Skills

**Generative AI & LLMs**: Anthropic Claude API, OpenAI API, Hugging Face (Transformers, Hub models), LLM integration, prompt engineering & versioning, structured outputs, context engineering, model routing, prompt caching, token & cost optimisation

**Agentic AI**: agentic workflows, multi-stage AI pipelines, tool and function calling, LangGraph, LangChain, MCP (Model Context Protocol), MCP tool integrations, Claude Code, OpenAI Codex, agent memory, multi-agent orchestration, human-in-the-loop

**RAG & Retrieval**: RAG, embeddings, vector search, vector databases, semantic search, hybrid retrieval, reranking, chunking & indexing, grounded generation

**AI Reliability & Evaluation**: schema validation & retry, output guardrails, automated QA gates, eval harnesses, retrieval evaluation, LLM observability, prompt and model evaluation, failure analysis

**Computer Vision**: OpenCV, YOLO, object detection, object tracking, real-time video processing

**AI Integrations & Automation**: Apify, Aircscale, MillionVerifier, Instantly, third-party APIs, webhooks, lead-enrichment pipelines, automated outreach workflows, data extraction and processing

**Backend**: Python 3.12+, FastAPI, async and await, asyncio, Pydantic v2, SQLAlchemy 2.0 async, PostgreSQL, WebSockets, background jobs, REST APIs, API authentication

**Frontend**: React 18 and 19, TypeScript, Next.js, Vite, TanStack Start, Router and Query, Tailwind CSS, Radix UI and shadcn, Framer Motion, React Three Fiber, React Hook Form, Zod

**DevOps & Production**: Docker, Vercel, Railway, AWS S3, Cloudflare, GitHub and GitLab, CI/CD, production monitoring, pytest

## Professional Experience

### Intagleo Systems

Jul 2024 – Present

Software Engineer (Full Stack, Python)

Lahore, Pakistan

- Build **LLM-powered product features on the Anthropic Claude API**: agentic workflows, RAG pipelines, MCP-connected tooling and prompt-engineered integrations, with structured outputs, schema validation and deterministic fallbacks so AI features fail safely in production.
- Ship secure, scalable **REST APIs in Python FastAPI** across several production applications, covering authentication, Pydantic validation, background jobs and business logic, plus Django and DRF services with role-based permissions.
- Design and optimise data across **PostgreSQL**, MySQL and MongoDB: schema design, efficient queries, migrations and data integrity across systems.
- Deliver full-stack features with **React and Next.js** front ends against FastAPI services, and real-time **WebSocket** features for live chat and notifications.
- Automate internal engineering workflows (task routing, approvals, compliance reporting) in Python, and deliver in **Agile** teams with Git and GitHub code review, sprint planning and weekly client meetings.

### Devsloop

Jul 2023 – Jun 2024

Associate Software Engineer

Lahore, Pakistan

- Delivered full-stack features on client web applications, **Python FastAPI** and Django REST endpoints with **React** front ends, from ticket through code review to deployment in a small Agile team, including **MySQL and PostgreSQL** schema changes, query tuning and bug triage.

### Wisdom Software House

Jan 2023 – Jun 2023

Software Developer Intern

Lahore, Pakistan

- Built full-stack web applications with a **Python FastAPI** backend and React front end, and optimised **MySQL** queries for performance during a six-month internship.

## Education & Certifications

**University of the Punjab, Lahore**, BSc Computer Science (BSCS)

2020 – 2024

**Certifications**: Udemy: Python for FastAPI: Advanced Backend Development · Python for Computer Vision & Deep Learning

**Awards**: Battle Byte Programming Competition, Runner-Up (university level)

**Languages**: English (fluent) · Urdu (native) · Punjabi (native) · **Driving Licence**: Pakistan (LTV)

References available on request

## Projects

AI Systems: self-built products · full write-ups at [arslantabish.com/#projects](https://arslantabish.com/#projects)

### Cold Outreach Pipeline: AI Lead-Generation SaaS [Live](#) · [Code](#)

Python, FastAPI, Claude, PostgreSQL, Stripe

- Designed a **9-stage, status-driven, resumable AI pipeline** (extract, enrich, AI filter, gate, email, verify, AI copy, QA, sync) where every lead advances only on a valid stage transition, so runs crash and resume without duplicating work or LLM spend.
- Built a defensive **structured-output layer** for all AI calls: fence stripping, JSON recovery, schema validation and retry with a corrective nudge before a typed `ParseError`, plus deterministic guardrails and an **automated QA gate** that blocks weak copy.
- Integrated **Apify, Aircscale, MillionVerifier and Instantly** into a single enrichment-and-outreach flow with webhook-driven status sync and per-provider fallbacks.
- Implemented multi-tenant isolation, BYOK keys encrypted at rest (AES-256-GCM), Stripe billing with plan limits, **per-call token-cost attribution**, and a zero-spend demo mode using a mock LLM behind a Python Protocol.

### DocPilot: Enterprise RAG over Private Documents [Code](#)

Python, FastAPI, PostgreSQL + pgvector, Claude, Docker

- Built a **permission-aware knowledge assistant** on hybrid retrieval: dense pgvector HNSW search in parallel with Postgres full-text search, Reciprocal Rank Fusion, cross-encoder reranking and small-to-big section expansion, with tenant and ACL predicates **compiled into the retrieval SQL** and row-level security so search can never return a chunk the asker cannot read.
- Grounded generation with **citation markers validated against real chunk ids** (invented markers stripped and logged), confidence blended from rerank strength and citation coverage, and a first-class `INSUFFICIENT_EVIDENCE` abstention path.
- Layout-aware ingestion (PDF, DOCX, HTML, XLSX, PPTX) with per-page OCR fallback and clause-safe chunking; a **golden-set eval gate in CI** (recall, nDCG, faithfulness, citation precision, abstention) fails merges on a two-point regression; 82 unit tests.

### Dukandar AI: Voice-First Conversational Shopping Agent [Code](#)

Python, FastAPI, Claude, SQLAlchemy, React

- Built a **catalogue-aware conversational agent** that walks shoppers through a facet-driven dialogue (category → kind → colour → size → price) and mirrors its filters live onto the storefront grid.
- Constrained Claude slot extraction to facet values that exist in the live catalogue, **eliminating hallucinated and out-of-stock suggestions**, with a deterministic keyword parser as a \$0 fallback on missing key, timeout or malformed JSON.
- Built the session engine, faceted search, a TTS proxy for voice output and the storefront front end on async **SQLAlchemy 2.0 + PostgreSQL**.

### GoldWatch: AI Gold (XAUUSD) Trading Analyst [private repo, walkthrough on request](#)

Python, Claude, market-data APIs

- Built a market-analysis assistant that computes market structure and indicators, renders charts, gathers macro context and asks Claude for a read, with one hard safety rail: it never places an order, only prints a full briefing (entry, stop, target, lot size, risk) for the human.
- Architected the system to **resist LLM over-eagerness**: a deterministic scanner scores setups first and an API call is spent only above a configurable threshold, keeping cost near zero on quiet days.

### Full-Stack & Frontend: client delivery

#### Khaayaal: Elder-Care Platform [Live](#) · [Code](#)

TanStack Start, React 19, TypeScript, Tailwind v4

- Designed and shipped the complete six-route marketing site on **TanStack Start + Router** with React 19 and Tailwind CSS v4, on a custom design-token system, Radix UI primitives and a reusable brand component library.
- Implemented an animated **3D hero with React Three Fiber**, scene-based scroll animation and an interactive pricing calculator over tiered care plans; React Hook Form + Zod, deployed on Vercel.

#### Leads In The Pipe: B2B Agency Website [Live](#) · [Code](#)

React 18, TypeScript, Tailwind, Framer Motion, FastAPI

- Translated a Figma design into a production single-page front end on Vite + React 18 + TypeScript (strict) with Tailwind design tokens and Framer Motion, backed by a Python FastAPI serverless API for form handling on Vercel.
- Enforced quality with Vitest + React Testing Library + vitest-axe, a **WCAG 2.1 AA** pass and SEO work (meta and OG tags, JSON-LD, Core Web Vitals).

### Professional Projects: Intagleo Systems (client and product names withheld; internal systems, not public)

#### Internal Engineering Operations Portal

FastAPI, PostgreSQL, GitHub API

- Built the full **FastAPI backend** for project tracking, developer task management and automated workflow processing: multi-department access with role-based permissions, real-time status updates, automated code-review routing and approvals via GitHub integration, and Excel compliance reporting.

#### Admin Panel Backend, enterprise client

Django, Django REST Framework

- Built secure REST APIs with **Django and DRF** (ORM, CBVs and FBVs) for complex business logic and workflow automation, with role-based access control, file uploads and automated email; unit tests, ORM query optimisation and weekly client meetings.

#### Multi-Database Service Platform

FastAPI, MongoDB, MySQL, AWS S3

- Developed a layered FastAPI service (routes, controllers, services and repositories) over MongoDB (Motor) and MySQL (aiomysql), with **TOTP two-factor authentication** (pyotp + QR provisioning), transactional email, AWS S3 file handling and structured logging.

#### Digital Signage Layout & Live-Preview Tooling

Node.js, Express, WebSocket, MCP

- Contributed to a layout and live-preview workspace for data-driven menu boards deployed by **national quick-service restaurant chains**: an Express + WebSocket dev server hot-reloading the board in an iframe, with data-bound 1920×1080 and 1080×1920 layouts joining HTML and CSS to spreadsheet data.
- Built an **agentic design-to-layout workflow over MCP**: connected Atlassian (Jira), Figma and Chrome DevTools MCP servers so an agent reads the ticket, pulls geometry and assets from the design, generates the board and verifies it pixel-for-pixel in the browser, turning a multi-day manual build into a reviewable pass.